

INHA UNIVERSITY IN TASHKENT (FALL 2024)

Computer Algorithms (CIE3090)

SAMPLE SOLUTION ASSIGNMENT – #3

Q1. Convert the following sequence to a Heap using Build Max Heap

12, 9, 7, 2, 15, 32

Use this space for showing iterations For I → 2 to 0		0	1	2	3	4	5		
		12	9	7	2	15	32		
i=2 L=5 R=6	i=2			i ←	→ L			Non leaf nodes	(n/2)-1 to 0
		12	9	32	2	15	7		
i=1 L=3 R=4	i=1		i ←	→ L		R		i=3 exit	
		12	15	32	2	9	7		
i=0 L=1 R=2	i=0	i ←	→ L		R			i=2 L=5 R=6	exit
		32	15	12	2	9	7		
				i			L		
		32	15	12	2	9	7		
		[1.5 Marks]							
1 mark for correct solution 0.5 mark for specifying i, L , R and showing arrows for swap									

Q2. Apply Quicksort Algorithm to sort the given sequence

(Show only steps of first PARTITION on the initial array)

Use this space for any calculations		1	2	3	4	5	6
P=1 r=6, x=10 initially J=0 I=-1		9	7	2	15	32	10
		A[j]< x i=j					
			A[j]< x i=j				
				A[j]< x i=j	j	j	
					i		
		9	7	2	10	32	15
		[1.5 Marks]					
		1 mark for correct solution 0.5 mark for explanation / steps					

Q2. Apply Merge sort Algorithm to sort the given sequence
(Show all steps)

Use this space for any calculations		1	2	3	4	5	6
		9	7	2	15	32	10
p=1	r=6	9	7	2			
q=3		9	7		15	32	10
				2	15	32	
		9					10
			7		15		
		7	9			32	
		2	7	9	15	32	
					10	15	32
		2	7	9	10	15	32

[1 Marks]

0.5 mark for correct solution

0.5 mark for explanation / steps